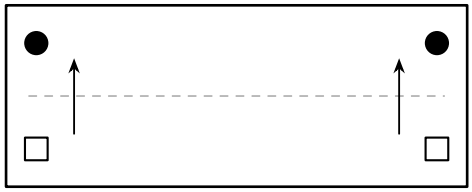
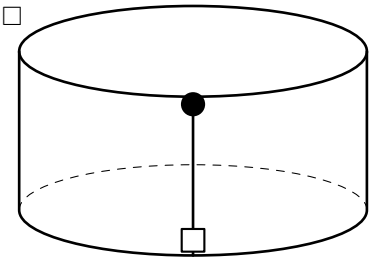
Keep the bands on the table. An adult makes every cut.

input process output



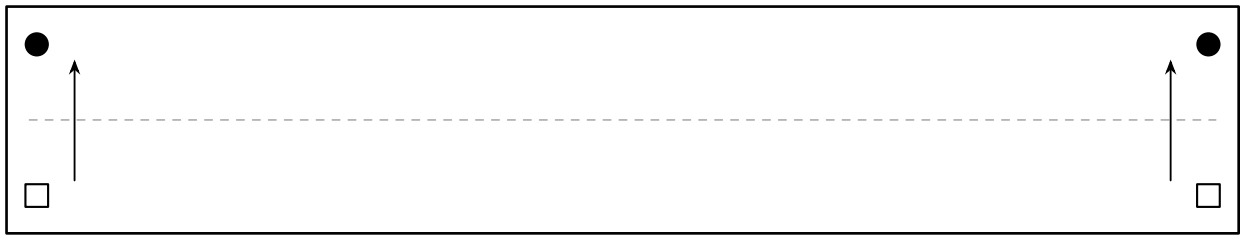
Join • to • and □ to □.



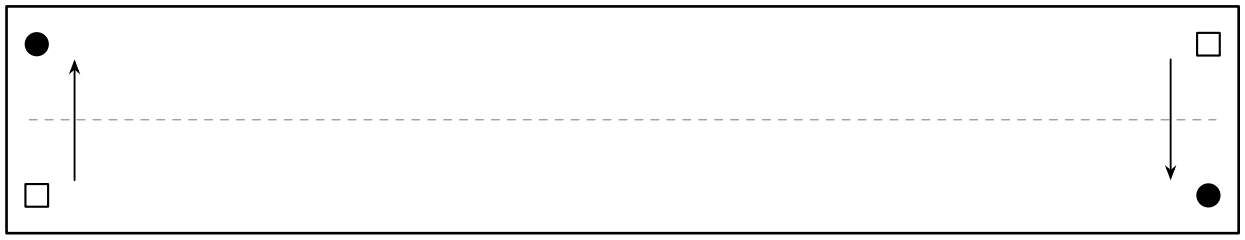
An edge trip stays on an edge, follows its seam marks, and stops on its first return to the start dot.

Problem 1: Trace every edge of A and B. Give each separate closed edge its own color.

A

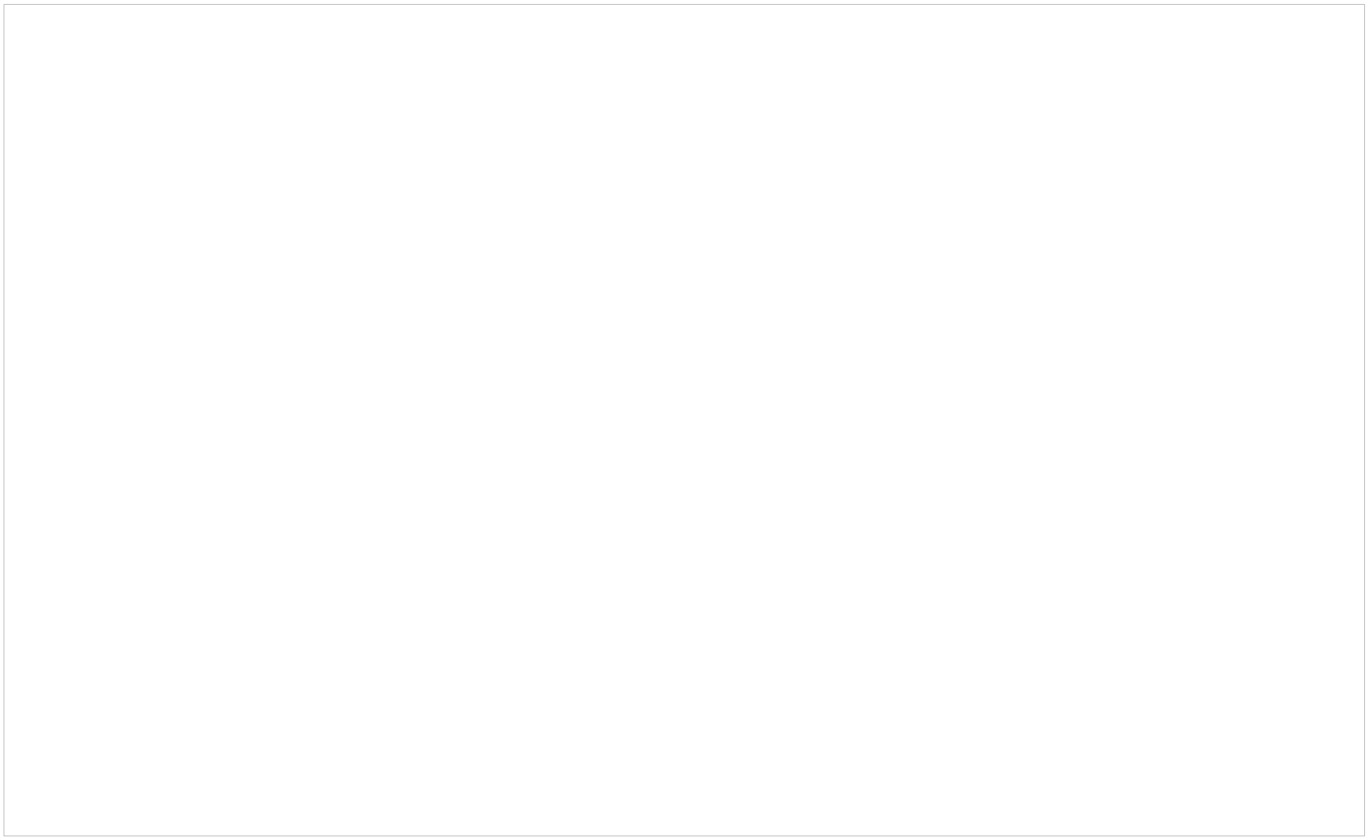


B

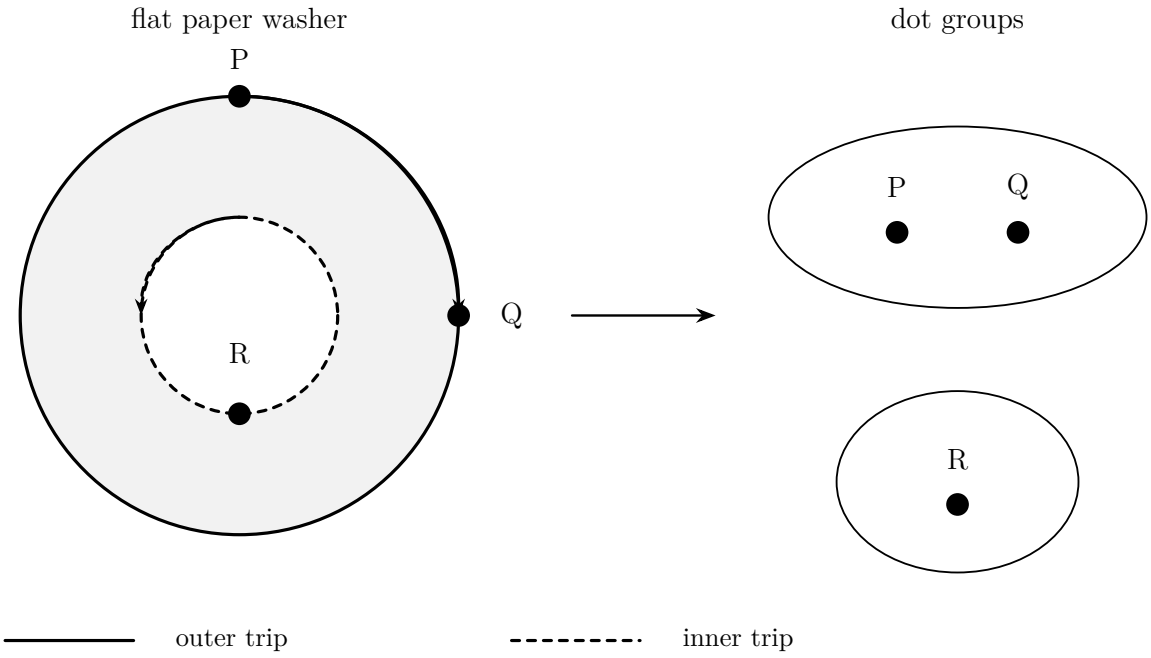


A: _____ B: _____

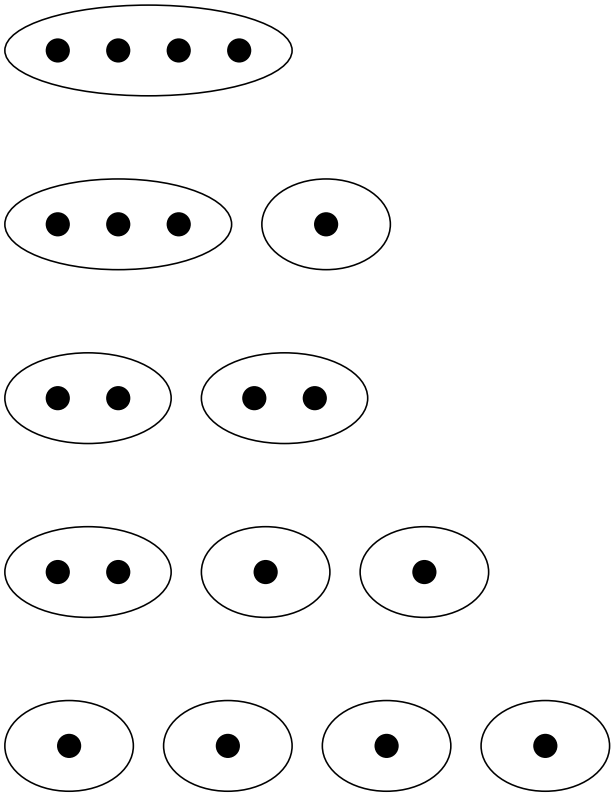
Problem 2: Put two dots on the edges of A so one edge trip reaches both, then try to place them so no one trip reaches both. Repeat with B.



Dots on one edge go in one ring.



Problem 3: Place four edge dots to match each picture on A and on B. Write yes or no in each blank. Each ring means a different closed edge; other edges may have no dots.



A: ____ B: ____

A: ____ B: ____

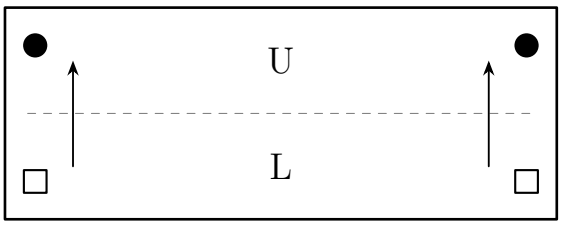
A: ____ B: ____

A: ____ B: ____

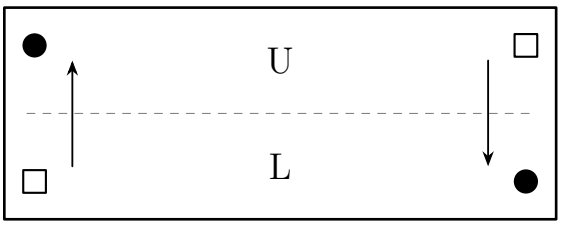
A: ____ B: ____

Problem 4: Guess how many pieces each band will make when an adult cuts its middle line. Ask the adult to cut A and B, and draw what you get.

A



B



I predict: _____

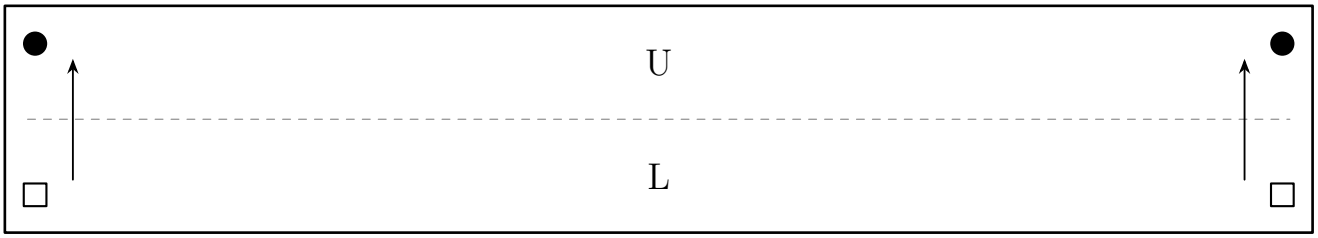
I predict: _____

after cutting A

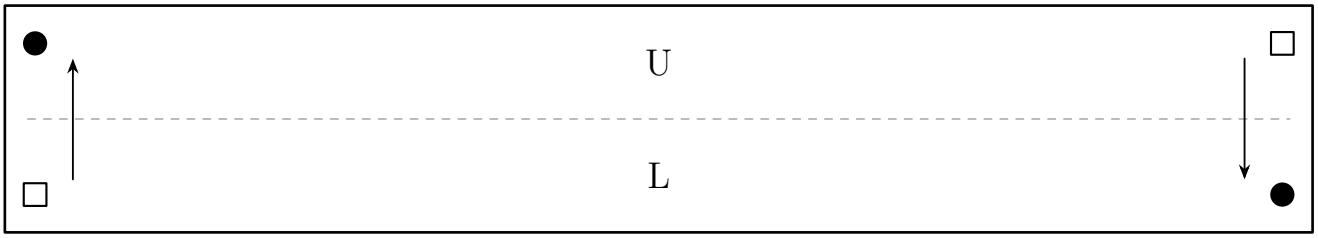
after cutting B

Problem 5: Trace every edge of each cut piece. Can two dots be on the same piece but on different closed edges?

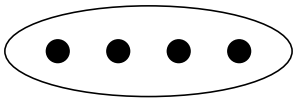
A



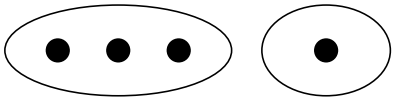
B



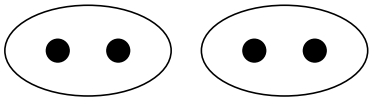
Problem 6: Use all the cut pieces from A together, then all those from B. Which four-dot pictures from Problem 3 can you now make? Write yes or no in each blank.



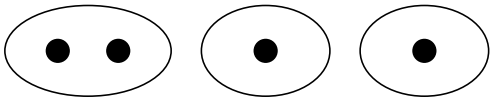
A: _____ B: _____



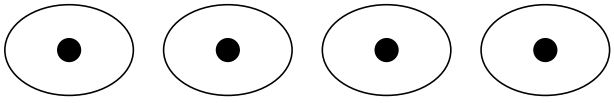
A: _____ B: _____



A: _____ B: _____



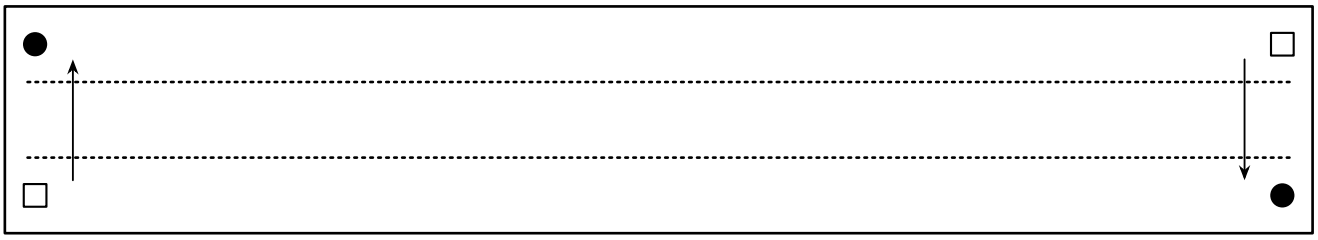
A: _____ B: _____

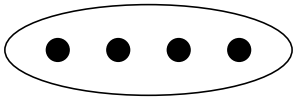


A: _____ B: _____

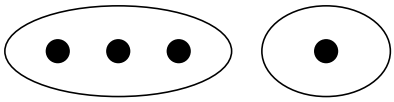
Problem 7: An adult cuts a new B one third of the way across, going round until the cut closes. Which four-dot pictures can you make now? Write yes or no in each blank.

B

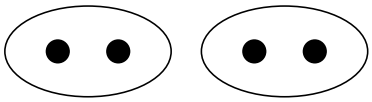




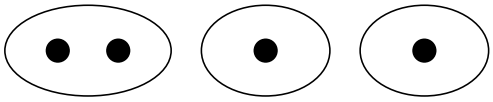
new B: _____



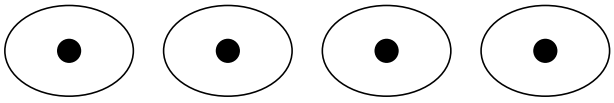
new B: _____



new B: _____



new B: _____



new B: _____